

4(a)	<i>any three from</i> protein coat / capsid ; I protein layer nucleic acid ; DNA <u>or</u> RNA ; R in nucleus I in cytoplasm I <i>ref. to</i>, double / single, strand acellular / not made of cells ; I absence of named cell structures AVP ; e.g. <i>ref. to</i> capsomeres size 15 nm to 1000 nm <i>accept any in range</i> (some) are enveloped / have phospholipid bilayer I with glycoproteins	3
4(b)(i)	one value in the range 64% – 68% ; $((940\,000 - 980\,000) \div (1\,440\,000 - 1\,480\,000)) \times 100$	1
4(b)(ii)	<i>allow women for pregnant women and therapy / treatment, for ART</i> <i>any three from</i> 1 (slight) decrease in (total) number of women living with HIV <u>and</u> (overall) increase in number of women living with HIV receiving ART ; 2 slight decrease / plateau / AW, between 2009 to 2010, in number of women living with HIV receiving, ART / therapy ; 3 proportion / percentage cover(age), of women receiving ART increases (in time period) ; A calculated values (approx. 13% to 66%) A number of women receiving ART increases more steeply than decrease in number of women living with HIV 4 data to support mp1 or 2 ; <i>mp1 two years and, two values / manipulated data, for either curve</i> <i>mp2 two values, 2009 compared to 2010 or manipulated data</i> <i>allow $\pm 20\,000$ for extracted values</i>	3

Question	Answer	Marks
4(b)(iii)	<i>in context of pregnant and breastfeeding women who are living with HIV any three from</i> 1 <i>ref. to mother to child transmission ; in context of HIV transmission</i> A (because) HIV can be passed from mother to baby A decreases HIV transmission during, pregnancy / labour / birth / breastfeeding A reduces, number / proportion, of babies born with HIV (so fewer die) I stops transmission (<i>as this is in context of global transmission this implies in all cases</i>) I makes babies immune to HIV / AW or gives passive immunity <i>allow idea that ART may be passed across, placenta / breastmilk, to baby and so provide(short-term) protection against any HIV transmitted from mother</i> 2 reduces number of, HIV positive women becoming ill (with HIV/AIDS) /women with HIV/AIDS dying from the disease ; A opportunistic infections / named examples e.g. TB 3 example of reduces spread of HIV ; <i>in any correct context other than HIV mother to child</i> e.g. child may grow up without HIV and will not pass on mother less likely to pass on to partner 4, 5 examples of, social / economic, effect ;; e.g. (healthy women) can contribute to work force can be main carer if partner has died (<i>idea that</i> children not orphaned) overall financial savings e.g. if infants are not born with HIV then no lifelong ART required ART may be less costly than treating HIV/AIDS makes breastfeeding safer when no other options exist to feed babies HIV negative children will become next workforce generation all women throughout world receive same treatment	3
4(c)(i)	(HIV) antigen / p24 ; A capsid protein / capsomere(s) / protein coat R HIV	1
4(c)(ii)	(time needed) so, <u>immune response</u> / clonal expansion / production of B-lymphocytes / production of plasma cells, can occur ; A B-cells / splenocytes R plasma cells need to multiply I <i>ref. to</i> antibody production	1

Question	Answer	Marks
4(c)(iii)	<i>any one from</i> immortal / long-lived ; able to replicate / capable of cell division ; uncontrolled cell division, can grow / survive, in cell culture ; cannot grow on, HAT / hypoxanthine-aminopterin-thymidine / step 4, (culture) medium ; A do not have gene coding for ability to grow on HAT	1
4(c)(iv)	<u>hybridoma</u> ;	1
4(c)(v)	<i>any one from</i> (check cells for) production / AW, (by hybridoma cells) of, anti-HIVp24 <u>antibody</u> / <u>antibody</u> against p24 ; A the antibody / monoclonal antibody A check cells, contain / have / AW, desired antibody / AW <i>idea that</i> only want cells that produce desired <u>antibody</u> / do not want cells that produce different antibodies / need to remove cells that don't produce the antibody ; waste of, money / resources, to culture other cells / if no antibody produced ;	1